

3.4 The threats to the tropical rainforest ecosystem and how they can be managed

- This chapter will explain:
- ★ the main threats to tropical rainforest ecosystems
 - ★ the impacts and causes of deforestation in the Brazilian Amazonian rainforest
 - ★ to what extent the strategies and techniques used to manage tropical rainforest ecosystems have been successful
 - ★ detailed specific example: Danum Valley Conservation Area, Borneo.

The main threats to tropical rainforest ecosystems

About 200 million people live in areas that are or were covered by tropical rainforests. These areas offer many advantages for human activities, such as farming, hydroelectric power, tourism, fishing and food supply, mineral development and forestry ([Figure 3.16](#) and [Table 3.4](#)). Rainforests also play a vital role in regulating the world’s climate, and they account for 50 per cent of the world’s plants and animals. They are vital too for the protection of soil and water resources.

▼ **Table 3.4** The value of tropical rainforests

Industrial uses	Ecological uses	Subsistence uses
● Charcoal ● Saw logs ● Gums, resins and oils ● Pulpwood ● Plywood and veneer ● Industrial chemicals ● Medicines ● Genes for crops ● Tourism	● Watershed protection ● Flood and landslide protection ● Soil erosion control ● Climate regulation, e.g. balancing levels of carbon dioxide and oxygen ● Special woods and ashes ● Fruit and nuts	● Fuelwood and charcoal ● Fodder for agriculture ● Building poles ● Pit sawing and saw milling ● Weaving materials and dyes ● Rearing silkworms and beekeeping



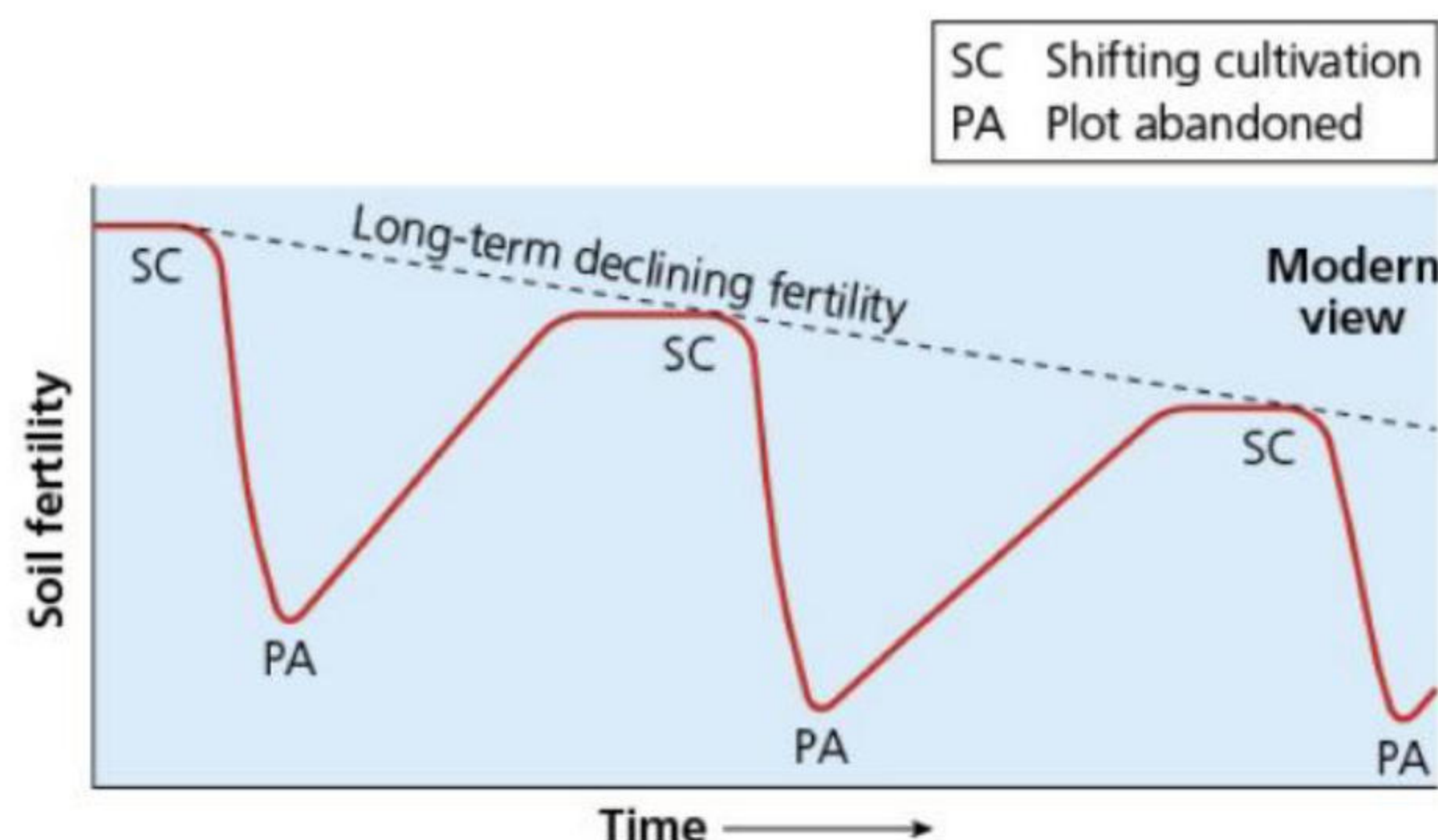
▲ **Figure 3.16** A tropical rainforest along with shifting cultivation – rice growing in Sarawak

However, tropical rainforests are under threat:

- » The year-round growing season is very attractive for farmers, although the poor quality of the soil results in the land being farmed for only a few years before it is abandoned ([Figure 3.17](#)). Nevertheless, large-scale **plantations** occur in areas of tropical rainforest, producing crops such as palm oil, which is increasingly being used for the **biofuels** industry.
- » Cattle ranching is a threat to rainforests as it replaces a multi-layered forest ecosystem, which intercepts most of the rainfall, with a single-layered grassland ecosystem, which gets trampled by cattle and can

produce much overland runoff and soil erosion.

- » The construction of roads and railways has two main impacts. The first is direct – rainforests are cut down to make way for roads, such as the Trans-Amazonian Highway. As a result of the increased accessibility provided by the new roads and railways, some settlers will cut down more rainforest to produce farmland.



▲ **Figure 3.17** The effect of rainforest clearance on long-term soil fertility

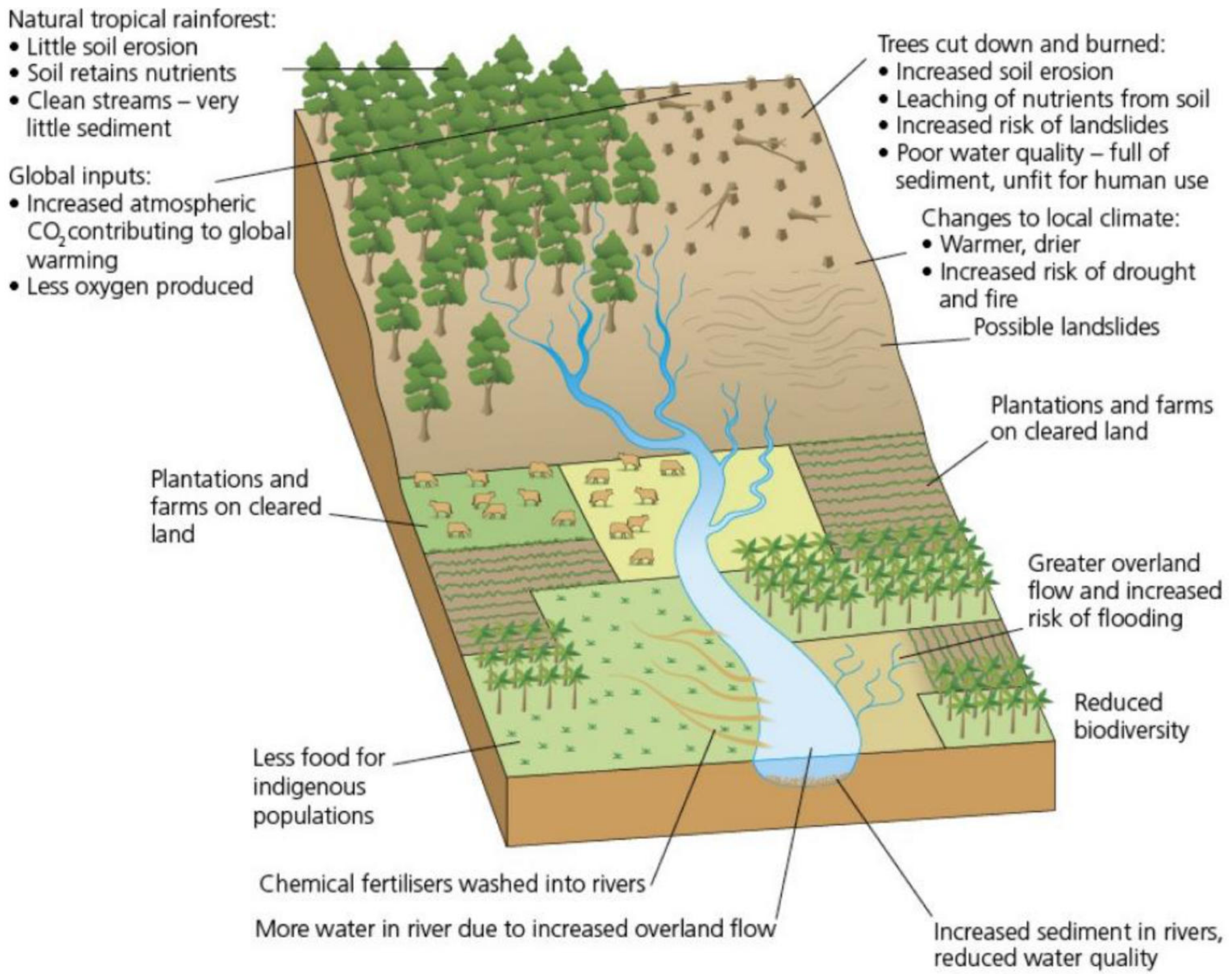
- » The second impact of the construction of roads and railways is that settlements will begin to develop along these roads and railways to provide more services for the new settlers. Each of these developments leads to a decline in the area of tropical rainforest and the services that it provides to people.
- » High rainfall totals, especially in hilly areas, favour the development of hydroelectric power (HEP), such as at Batang Ai in Sarawak, Malaysia ([Figure 3.19](#)).
- » Areas of rainforest have a long history of commercial logging. Tropical hardwoods, such as teak and mahogany, are prized by furniture manufacturers.
- » Mineral developments, such as iron ore at Carajas in Brazil and ilmenite on the southeast coast of Madagascar, are also developed in some rainforest areas and these resources may be exploited.

The impacts and causes of deforestation in the Brazilian Amazonian rainforest

Impacts

There are a large number of effects of deforestation ([Figure 3.18](#)), including:

- » disruption to the circulation and storage of nutrients
- » surface erosion and compaction of soils
- » increased flood levels and sediment content of rivers
- » climatic change
- » loss of biodiversity and animal habitats.



▲ **Figure 3.18** Some impacts of deforestation



▲ **Figure 3.19** The rainforest at Batang Ai, affected by flooding, shifting cultivation and soil erosion

Deforestation disrupts the closed system of **nutrient cycling** within tropical rainforests. Inorganic elements are released through burning and are quickly flushed out of the system by the high intensity rains.

Soil erosion is also associated with deforestation. As a result of soil compaction, there is a decrease in infiltration, an increase in overland runoff and surface erosion.

As a result of the intense surface runoff and soil erosion, rivers have a higher **flood peak** and a shorter time lag. However, in the dry season river levels are lower, the rivers have greater turbidity (murkiness due to more sediment), an increased bed load, and carry more silt and clay in suspension.

Other changes relate to **climate**. As deforestation progresses, there is a reduction of water that is re-evaporated from the vegetation, hence the recycling of water is diminished. Evaporation rates from savanna grasslands are estimated to be only about one-third of that of the tropical rainforest. Thus, mean annual rainfall is reduced, and the seasonality of rainfall increases.

Causes

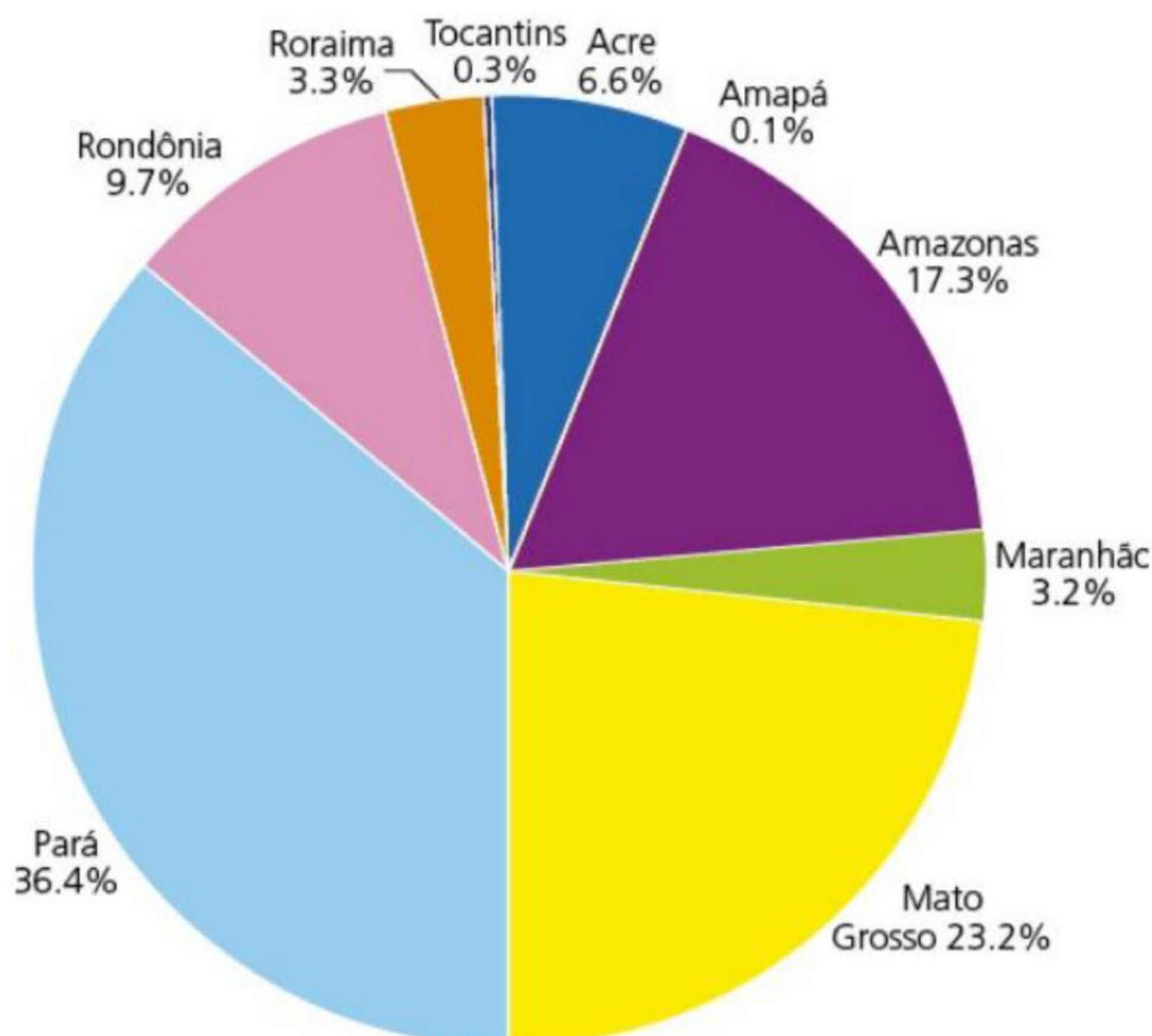
There are six main causes of deforestation in Brazil:

- » Agricultural colonisation by landless migrants and speculative developers along highways and agricultural growth areas.
- » Conversion of the forest to cattle pastures, especially in eastern and southeastern Para and northern Mato Grosso.
- » Mining, for example the Greater Carajas Project in southeastern Amazonia, which includes a 900 km railway and extensive deforestation to provide charcoal to smelt the iron ore; another threat from mining comes from the small-scale informal gold mines, *garimpeiros*, causing localised deforestation and contaminated water supplies.
- » Large-scale hydroelectric power schemes such as the Tucuruí Dam on the Tocantins River.
- » Forestry taking place in Para, Amazonas and northern Mato Grosso.
- » Use of the forest by local populations is a necessity in order to earn a living and have an acceptable quality of life.

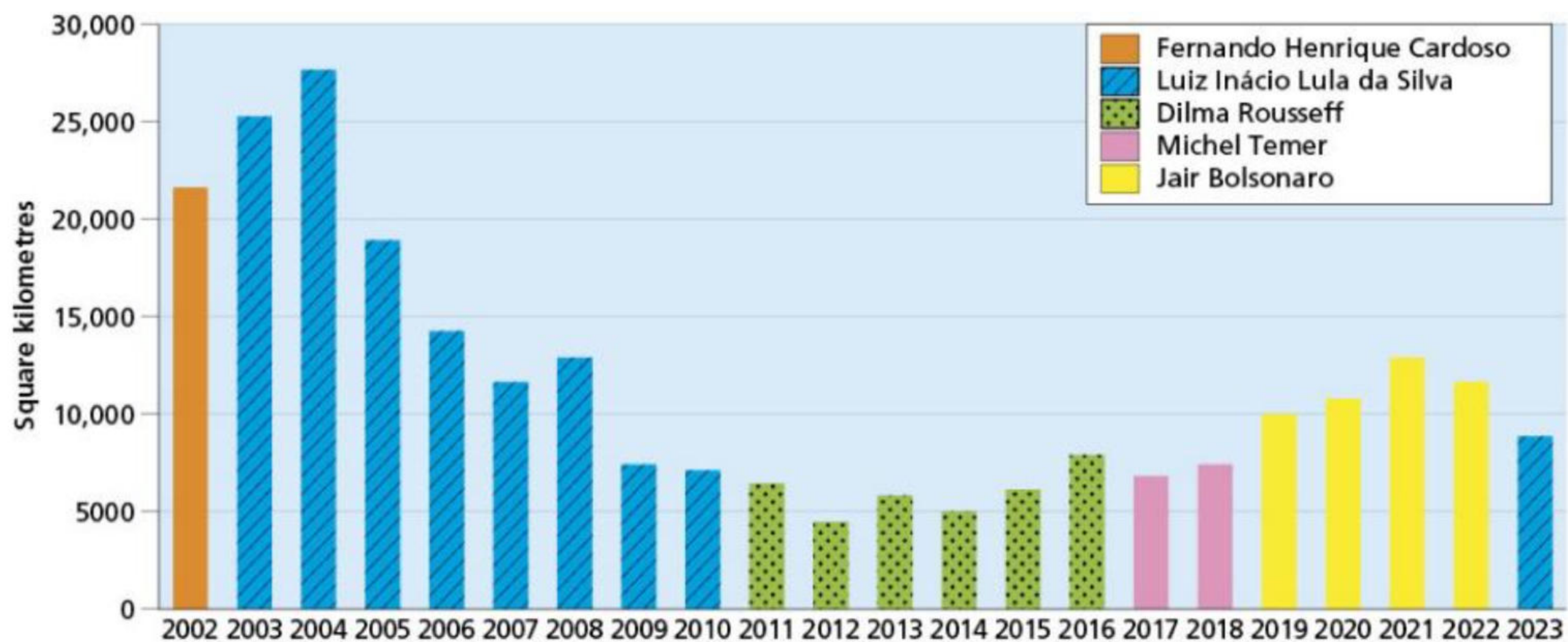
Deforestation in Brazil shows several trends:

- » It is a recent phenomenon.
- » It has partly been promoted by government policies.
- » There is a wide range of causes of the deforestation.
- » It includes new areas of deforestation as well as the extension of previously deforested areas.
- » A major cause of the deforestation is land speculation and the granting of land titles to those who 'occupy' parts of the rainforest.

According to the news organisation Mongabay, in 2023 deforestation from conversion of forest to soy farming continued to spread through Brazil's Cerrado and Amazon rainforest. There was around 27,000 ha of deforestation and forest degradation in the Cerrado in just four months, while the Amazon saw over 30,000 ha. All of it was located near grain silos used by the seven biggest soy traders in Brazil ([Figures 3.20](#) and [3.21](#)).



▲ **Figure 3.20** Deforestation of the Amazon by state, 2023



▲ **Figure 3.21** Deforestation of the Amazon under the president in office at the time, 2002–23

[Figure 3.20](#) shows that most of the deforestation occurred in just two states, Pará and Mato Grosso. However, the long-term trend in deforestation is going down. The current decline is largely due to Brazilian President Lula, who aims to eliminate illegal deforestation by 2030. This contrasts with the significant increase in deforestation under the previous president, Jair Bolsonaro.

Activities

- 1 Comment on the value of tropical rainforests for human populations.
- 2 Outline the main impacts of deforestation on the natural environment.
- 3 Explain the main causes of deforestation in Brazil.
- 4 Comment on the trends in deforestation in Brazil.

The strategies and techniques used to manage tropical rainforest ecosystems

Because tropical rainforests are important for many services, such as flood control, climate regulation, a source of genetic material, timber and food, and many others ([Table 3.4, page 80](#)), attempts have been made to protect rainforests. An example of the sustainable management of tropical rainforests is the Heart of Borneo (HoB) project. The governments of Malaysia, Indonesia and Brunei created the HoB, an area of over 20 million ha, to sustainably protect the rainforest and water resources for the welfare of present and future generations. It protects forest cover, manages protected areas and maintains biodiversity and environmental services in productive activities such as farming, forestry, fishing and ecotourism. Nevertheless, many areas of rainforest are still being cut down, in part due to population growth and the need to find more food supplies, but also due to people who want to cut down rainforests to get at mineral resources below ground.

Elsewhere in Borneo, over two-thirds of Borneo's dipterocarps (tall, straight trees providing high quality timber) are threatened with extinction. Up to 350 species of mammals, birds, reptiles and fish are at risk of extinction, for example clouded leopards, orangutans and Sumatran rhinoceroses. Other impacts include habitat losses, carbon emissions, fires (used to clear forests), land degradation and decreased water quality – palm oil plantations often use chemical fertilisers and pesticides which can pollute rivers and groundwater. Land drainage increases soil erosion. There are also social issues as the palm oil industry often removes indigenous people from their land.

Detailed specific example

Danum Valley Conservation Area, Malaysian Borneo

The Danum Valley Conservation Area (DVCA) in Borneo contains more than 120 mammal species, including ten species of primate. The DVCA and surrounding forest is an important reservation for orangutans. These forests are particularly rich in other large mammals, including the Asian elephant, Malayan sun bear, clouded leopard, bearded pig and several species of deer. The area also provides one of the last refuges in Sabah, northeast Borneo, for the critically endangered Sumatran rhino. Over 340 species of bird have been recorded at Danum, including the argus pheasant, Bulwer's pheasant, and seven species of pitta bird.

The DVCA covers 43,800 ha, comprising almost entirely lowland dipterocarp forest (dipterocarps are valuable hardwood trees). It is the largest expanse of pristine forest of this type remaining in Sabah ([Figure 3.22](#)).

Until the late 1980s, the area was under threat from commercial logging. The establishment of a long-term research programme between Yayasan Sabah and the Royal Society in the UK created local awareness of the conservation value of the area and provided important scientific information about the forest and what happens to it when it is disturbed through logging. Danum Valley is controlled by a management committee containing all the relevant local institutions –

the wildlife, forestry and commercial sectors are all represented. This allows Danum Valley to be managed sustainably. To the east of the DVCA is the 30,000 ha Innoprise-FACE Foundation Rainforest Rehabilitation Project (INFAPRO), one of the largest forest rehabilitation projects in Southeast Asia, which is replanting areas of heavily disturbed logged forest.

Because all areas of conservation and replanting are embedded within the larger commercial forest, the value of the whole area is greatly enhanced. Movement of animals between forest areas is enabled and allows the continued survival of some important and endangered animals such as the Sumatran rhino, the orangutan and the Borneo elephant.

In the late 1990s, a hotel was established on the northeastern edge of the DVCA. It has established flourishing ecotourism in the area and exposed this unique forest to a wider range of visitors than was previously possible. As well as raising revenue for the local area, it has raised the international profile of the region as an important centre for conservation and research. This is a very good example of the sustainable development of an area of tropical rainforest.

Activities

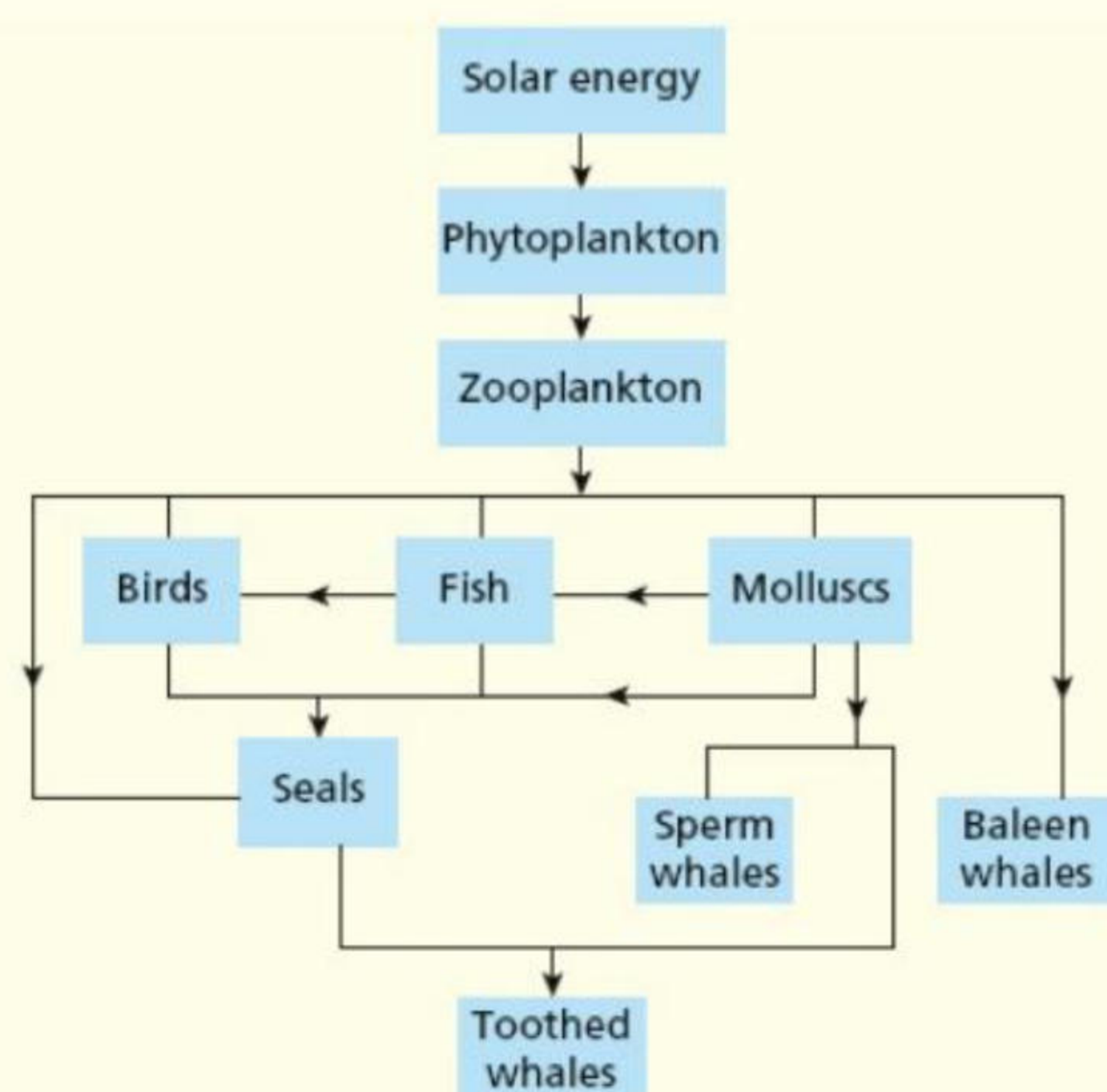
- 1 What was the main threat to the Danum Valley before the late 1980s?
- 2 Why is the DVCA important for the conservation of species?



▲ **Figure 3.22** The Danum Valley Conservation Area

Practice questions

- 1 [Figure 3.23](#) shows a food web for the sea around Antarctica.



▲ **Figure 3.23** A food web for the sea around Antarctica

- a i Identify the main source of energy for this ecosystem. [1]
 - ii Identify the organism that feeds on phytoplankton. [1]
 - iii State the organism that feeds on birds. [1]
 - iv State the top carnivore in the Antarctic marine ecosystem. [1]
 - b Explain why Antarctica's ecosystems are largely marine-based rather than land-based. [3]
- 2 View [Figure 3.16 on page 80](#) which shows an area of tropical rainforest.
- a Identify how this area of rainforest is being used. [1]
 - b Compare the characteristics of the vegetation in the rainforest with that of the cleared area. [2]
 - c Explain two reasons why tropical rainforests are being destroyed by human activities. [4]
 - d Explain the consequences of the deforestation of tropical rainforests. [4]